

MARYAM SARFARAZ KHAN

SENIOR HVAC DESIGN ENGINEER

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SKILLS AND CERTIFICATIONS

Member of HVACR Society	Cooling Load Calculations	3D Modeling
PEC Registered Engineer	MEP Emaar Guidelines	Product service
Certificate of VRF System	Energy Conservation	High-rise story
HVAC certification	Project Management Training	Testing
Certificate of Chiller System	Sales Processes & Engineering	HVAC Design
Certificate of ASHRAE Standards	Building Codes & Performance	ASHRAE 62.1
Certificate of Ventilation System	Engineering & Shop Drawings	System Design
Building Codes & Standards	Product Management Certificate	Roof Insulation

TECHNICAL EXPERTISE

HAP SOFTWARE (Cooling Loads + Energy Modelling also)	Pipe Sizers
EQUEST (Heating Loads + Energy Modelling)	Duct Sizers
IES (Basics of modelling)	VRF Sizing Software
AUTOCAD (Basics of Revit)	Bluebeam Software
SPC-HPD (Fresh Air Calculation)	Pump Head Calculation
Psychrometric Analysis	Bill of Quantities
Winergy (Enthalpy Wheel Efficiency and Temp)	Project Cost Estimate
Equipment Selection Software (FAHU, MAHU, AHU)	Equipment Schedule
Cost Estimate of the project	Compliance Sheet

EDUCATION

Bachelors in Mechanical Engineering (IMD)
NED University of Engineering and Technology,
Karachi



GENERAL RESPONSIBILITIES

Here are some of the key roles I have played in some of these working careers:

- ❖ Heating and cooling load calculations following ASHRAE standards using software (Carrier HAP). Select the HVAC equipment based on heating and cooling load calculations. Selection of diffusers, grills, and exhaust fans according to calculated CFM and TR. Calculation of duct sizes for all types of HVAC units. Psychrometric analysis for all types of Air handling units.
- ❖ Preparation of plans, details, specifications, BOQ and cost estimations of HVAC works.
- ❖ Review and check the shop drawings and material submittals.
- ❖ Ensure statutory and regulatory requirements are met in all designs.
- ❖ Support the project submittals and No Objection Certificates process to the relevant Authorities.

- ❖ Stay updated with relevant technical developments within the discipline.
- ❖ Understand the contract and scope of project deliverables.
- ❖ Ensure that Health & Safety is embedded into all work practices in line with company policies.
- ❖ Provide regular work status updates, including risks or concerns, to Senior or MEP manager.
- ❖ Attend and contribute to team and project meetings as required.
- ❖ Contribute to the delivery of the Building Division strategy. Including supporting the companies approach to sustainability.
- ❖ Ensure QMS and Project Life cycle compliance across team and projects.
- ❖ Complete accurate time sheets by set deadline.
- ❖ Feedback on all business development opportunities.
- ❖ Evaluated project feasibility based on material, equipment, and labour expenses.
- ❖ Addressed design challenges and evaluated alternative design models to meet project requirements.
- ❖ Brainstormed with product designers to implement and refine new manufacturing processes.

WORKING

Mechanical Design Engineer (2023 to Present)

CKR CONSULTING ENGINEERS



CKR Consulting Engineers has grown to become one of the leading multi-disciplinary engineering firms in Africa, the Middle East and Asia. CKR Consulting Engineers can address all your built environment engineering consultancy needs by providing services in the Electrical, Mechanical, Civil, Structural, High Voltage, ICT infrastructure and ELV fields.

Projects:

There is several projects covered in this tenure. Some of them are highlighted below as per the various building applications:

Wadi Safar Aman Hill Top Hotel KSA

The resort features numerous assets, positioned throughout an almost two-million-square-metre site, with a 78-key hotel, 29 branded villas and four branded farms featuring sweeping grounds and orchards embedded within the hilltops and mountains. The GFA is 128,357 m² and is located in Riyadh, Saudia.

DESIGN GUIDELINES

The designs have been developed and coordinated based on the following documentation, standards, codes and guidelines:

- Aman Design Guidelines for Branded Resorts & Residences Revision H – January 2021
- MEP Requirements – Electrical Systems –11 November 2019
- Mechanical & Electrical Requirements – Electrical Schedule - Indoor Design Conditions – 11 November 2019
- MEP Planning Guidelines - Mechanical & Electrical Requirements - HVAC –11 November 2019
- Mechanical & Electrical Requirements – HVAC Schedule - Indoor Design Conditions - 11 November 2019
- MEP Planning Guidelines - Mechanical & Electrical Requirements – Plumbing – 11 November 2019
- MEP Planning Guidelines - Information Technology Structured Cabling – Voice & Data – 21 December 2021
- Mechanical & Electrical Requirements – Building Management System (BMS) – 11 November 2019

- Diriyah Gate Master Plan and Plot Regulation Sheets
- Saudi Building Codes
- Ministry of Tourism
- DGDA Car Park Guidelines
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Emaar Beachfront Plot BF09, Dubai Harbour, Marina, Dubai

Emaar Beachfront BF09 is completely new experience for me. It is an outdoor cooling in its All-Day Dining Terrace & Pool Bar Terrace. It is based on 100% Outdoor Air. Airflow rate is Estimated as 35 l/s/m².

DIFC Mixed Use East, Dubai

DIFC mixed use is a great experience working on this project. It corresponds to around Four buildings. I have been working on two towers named as Carlyle Hotel & Four Seasons which corresponds to 32,000 m². Almost 1600 TR Cooling capacity required from District Cooling. It was comprised of two ETS Plantrooms on different levels on each tower. Each tower is based on multi-use application like residential, hotel, amenities etc. 50 l/s Chilled water Flowrate required for each primary/secondary pumps for each tower. FCU, MUAHU, OAHU, Fans & Ecology unit's calculations are done by me in this project.

WORK HISTORY

Sr. Mechanical Project Engineer (2017 to 2023) (Working remotely for UAE office)

[S. MEHBOOB & COMPANY CONSULTING ENGINEERS](#)



SMC provides excellence in engineering services to our clients within their budgetary and Technological Constraints. SMC is a firm that is abreast of the latest technology, it is a firm which is multidisciplinary with committed principals.

Projects:

There is a huge list of projects covered in this tenure. Some of them are highlighted below as per the various building applications:

➤ EMAAR CRESCENT BAY

Emaar is a huge project it has number of towers located in Defense Phase 8, Clifton. It is a multi-use building. VRF system has been installed in all over the tower. Cooling load calculation, Fresh air calculations, Ventilation Calculations etc. and selection of units is done by me. Standards are used as per the Emaar Guidelines.

➤ YOUNIS VILLA & HAMMAD VILLA, DUBAI

It is a residential project adjacent to each other. VRF system has been installed in the complete building.

➤ CARREFOUR, OMAN

It is huge shopping mall located in Oman. It is based on ASHRAE 62.1 Standards of Ventilation. Air-cooled Chillers have been installed for Air-conditioning & heating in this project.

➤ SUNCOR ENERGY, CANADA

It is a project based on Canada. Its Energy modelling is done on Equest Software. Basic Heating is done in this project as per NECB Codes and Standards of Canada.

➤ BANK AL HABIB

Chiller is installed for this Bank. It was very difficult to pass through the ducts from limited spaces but ultimately it became a huge success by revolving and cutting edges of ductwork in the limited space and time. Energy modelling is also performed for this project on EDGE software, and it came out to be 33 %.

➤ TECHNOLOGY PARK, KARACHI

It is one of the huge projects which have total cooling load of 2000 Tons and Chillers have been installed for this project. Air cooled chillers and Cooling towers have been selected for this project. It is a triple round towers standing in a trio shape. Shades of other towers have been considered while calculating cooling loads on HAP.

➤ SHOKAT KHANUM HOSPITAL

It is one of the huge projects followed by Health care standards. Humidity levels and Temperatures have been considered as per the codes. Air changes have been considered for this project as per the healthcare facilities ASHRAE STANDARD 170-2017, Ventilation of Health Care Facilities.

HVAC Design Engineer (2014 to 2017)

[SEM ENGINEERS](#)



SEM Engineers is also one of the leading HVAC consultancies in Pakistan. It is an amazing experience working with multiple ranges of projects. Some of them are mentioned below:

Projects:

➤ REHMAN MEDICAL INSTITUTE OF DENTISTRY, PESHAWAR

RMI is an institute of dentistry, consists of 50,000 sq. feet. Designed the Air-conditioning system based on water cooled screw chiller system. It is designed based on ASHRAE standards.

➤ MCDONALD'S

McDonald's is a huge chain. Designed its Air-conditioning based on McDonald's standards. It has its own analysis for ventilation based on the Air-balance. Designed McDonald's of around many of the cities which include Round about located in Islamabad project based on 14000 sq. feet.

➤ DOMINO'S PIZZA

It is located in Karachi. Designed on the basis of Air balance standard method. Almost 5000 sq. feet.

➤ CANCER FOUNDATION HOSPITAL

CFH is a charity-based hospital. Its area is around 25,000 sq. feet. I have designed its air conditioning based on air change and ASHRAE recommended standards.

Mechanical Design Engineer (2012 to 2014)

A.R.T ENGINEERING



ART Engineering deals with HVAC projects within Pakistan. The Mega project which I have completed here is WTC (World Trade Centre Islamabad).

Projects:

➤ WORLD TRADE CENTRE, ISLAMABAD

A nine storied offices/shopping mall building located in Islamabad. Designed the Air- conditioning, Mechanical Ventilation system based on chiller system.

➤ SAIMA JINNAH, KARACHI

Saima Jinnah is a shopping mall near Defence Phase II, Karachi. It consists of 352,000 sq. feet area.